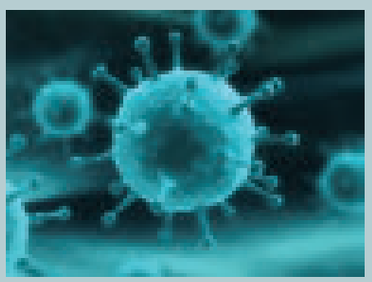
**Nipah virus** is a zoonotic virus (transmitted from animals to humans) and also transmitted through contaminated food. In infected people, it causes a range of illness from asymptomatic infection to acute respiratory illness and fatal encephalitis.



i. Pneumotropic diseases (respiratory tract infected by influenza)

ii. Dermotropic diseases (skin and subcutaneous tissues affected by chicken pox and measles)

iii. Viscerotropic diseases (blood and visceral organs affected by yellow fever and dengue fever)

iv. Neurotropic diseases (central nervous system affected by rabies and polio). Some common viral diseases of human beings are given in **table 7.2**.




Swine flu was first recognised in the 1919 pandemic and still circulates as a seasonal flu virus. Swine flu is caused by the H1N1 virus strain. Symptoms include fever, cough, sore throat, chills, weakness and body aches. Children, pregnant women and the elderly are at risk from severe infection.

## 7.1.2 Protozoan diseases

About 15 genera of protozoans live as parasites within the human body and cause diseases.

**Amoebiasis** also called amoebic dysentery or amoebic colitis is caused by *Entamoeba histolytica,* which lives in the human large intestine and feeds on mucus and bacteria **(Fig. 7.1)**. Infective stage of this parasite is the **trophozoite**, which penetrates the walls of the host intestine (colon) and secretes histolytic enzymes causing ulceration, bleeding, abdominal pain and stools with excess mucus. Symptoms of amoebiasis can range from diarrhoea to dysentery with blood and mucus in the stool. **House flies** (*Musca domestica*) acts as a carrier for transmitting the parasite from contaminated faeces and water.

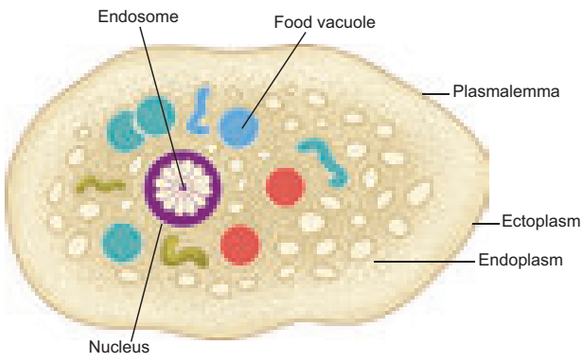



**Fig. 7.1 *Entamoeba histolytica***

**African sleeping sickness** is caused by *Trypanosoma species***.** *Trypanosoma* is generally transmitted by the blood sucking **Tsetse** flies. Three species of *Trypanosoma* cause sleeping sickness in man.

1. *T. gambiense* is transmitted by *Glossina palpalis* **(Tsetse fly)** and causes Gambian or Central African sleeping sickness **(Fig. 7.2).**
2. *T. rhodesiense* is transmitted by *Glossina morsitans* causing **Rhodesian or East African sleeping sickness.**
3. *T. cruzi* is transmitted by a bug called *Triatoma megista* and causes **Chagas disease or American trypanosomiasis.**

**Kala – azar** or visceral leishmaniasis is caused by *Leishmania donovani,* which is transmitted by the vector *Phlebotomus* **(sand fly)**. Infection may occur in the endothelial cells, bone marrow, liver, lymph glands and



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